Practical 3

```
# Install pgmpy if it's not already installed
!pip install pgmpy pandas numpy

import pandas as pd
from pgmpy.models import DiscreteBayesianNetwork
from pgmpy.estimators import MaximumLikelihoodEstimator
from pgmpy.inference import VariableElimination

# Load dataset
df = pd.read_csv("/content/heart.csv")

print(df.head())

# Define Network Structure
model = DiscreteBayesianNetwork([
    ('cp', 'target'),
    ('thal', 'target'),
    ('ca', 'target'),
    ('oldpeak', 'target')
])

# Learn Parameters
print('Learning CPD using Maximum likelihood estimators...')
model.fit(df) # Removed estimator argument to let it default to
DiscreteMLE()

print("Model Learned Successfully")

# Inference
infer = VariableElimination(model)

q = infer.query(
    variables=['target'],
    evidence={'cp': 2}
)

print(q)

# Computing the Probability of Heart Disease given chest pain type
print('\n2. Probability of HeartDisease given evidence= cp')
q2 = infer.query(variables=['target'], evidence={'cp': 2})
print(q2)
```

```
Requirement already satisfied: pgmpy in
/usr/local/lib/python3.13/dist-packages (1.1.2)
Requirement already satisfied: pandas in
```

```
/usr/local/lib/python3.13/dist-packages (2.2.3)
Requirement already satisfied: numpy in
/usr/local/lib/python3.13/dist-packages (2.1.3)
Requirement already satisfied: huggingface_hub>=0.23 in
/usr/local/lib/python3.13/dist-packages (from pgmpy) (1.29.0)
Requirement already satisfied: networkx>=3.0 in
/usr/local/lib/python3.13/dist-packages (from pgmpy) (3.6.1)
Requirement already satisfied: scipy>=1.10 in
/usr/local/lib/python3.13/dist-packages (from pgmpy) (1.16.3)
Requirement already satisfied: scikit-learn>=1.2 in
/usr/local/lib/python3.13/dist-packages (from pgmpy) (1.6.1)
Requirement already satisfied: statsmodels>=0.14.5 in
/usr/local/lib/python3.13/dist-packages (from pgmpy) (0.15.0)
Requirement already satisfied: tqdm>=4.64 in
/usr/local/lib/python3.13/dist-packages (from pgmpy) (4.67.3)
Requirement already satisfied: pyparsing>=3.0 in
/usr/local/lib/python3.13/dist-packages (from pgmpy) (3.3.2)
Requirement already satisfied: joblib>=1.2 in
/usr/local/lib/python3.13/dist-packages (from pgmpy) (1.6.0)
Requirement already satisfied: opt_einsum>=3.3 in
/usr/local/lib/python3.13/dist-packages (from pgmpy) (3.4.0)
Requirement already satisfied: scikit-base>=0.12.4 in
/usr/local/lib/python3.13/dist-packages (from pgmpy) (1.1.1)
Requirement already satisfied: python-dateutil>=2.8.2 in
/usr/local/lib/python3.13/dist-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in
/usr/local/lib/python3.13/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in
/usr/local/lib/python3.13/dist-packages (from pandas) (2026.3)
Requirement already satisfied: click<9.0.0,>=8.4.2 in
/usr/local/lib/python3.13/dist-packages (from huggingface_hub>=0.23-
>pgmpy) (8.5.0)
Requirement already satisfied: filelock>=3.10.0 in
/usr/local/lib/python3.13/dist-packages (from huggingface_hub>=0.23-
>pgmpy) (3.32.5)
Requirement already satisfied: fsspec>=2023.5.0 in
/usr/local/lib/python3.13/dist-packages (from huggingface_hub>=0.23-
>pgmpy) (2025.12.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.5.2 in
/usr/local/lib/python3.13/dist-packages (from huggingface_hub>=0.23-
>pgmpy) (1.6.0)
Requirement already satisfied: httpx<1,>=0.23.0 in
/usr/local/lib/python3.13/dist-packages (from huggingface_hub>=0.23-
>pgmpy) (0.28.1)
Requirement already satisfied: packaging>=20.9 in
/usr/local/lib/python3.13/dist-packages (from huggingface_hub>=0.23-
>pgmpy) (26.3)
Requirement already satisfied: pyyaml>=5.1 in
/usr/local/lib/python3.13/dist-packages (from huggingface_hub>=0.23-
```

```
>pgmpy) (6.0.3)
Requirement already satisfied: typing-extensions>=4.1.0 in
/usr/local/lib/python3.13/dist-packages (from huggingface_hub>=0.23-
>pgmpy) (4.16.0)
Requirement already satisfied: cloudpickle>=3.0 in
/usr/local/lib/python3.13/dist-packages (from joblib>=1.2->pgmpy)
(3.1.2)
Requirement already satisfied: six>=1.5 in
/usr/local/lib/python3.13/dist-packages (from python-dateutil>=2.8.2-
>pandas) (1.17.0)
Requirement already satisfied: threadpoolctl>=3.1.0 in
/usr/local/lib/python3.13/dist-packages (from scikit-learn>=1.2-
>pgmpy) (3.6.0)
Requirement already satisfied: patsy>=0.5.6 in
/usr/local/lib/python3.13/dist-packages (from statsmodels>=0.14.5-
>pgmpy) (1.0.3)
Requirement already satisfied: formulaic>=1.1.0 in
/usr/local/lib/python3.13/dist-packages (from statsmodels>=0.14.5-
>pgmpy) (1.2.2)
Requirement already satisfied: interface-meta>=1.2.0 in
/usr/local/lib/python3.13/dist-packages (from formulaic>=1.1.0-
>statsmodels>=0.14.5->pgmpy) (2.0.1)
Requirement already satisfied: narwhals>=1.17 in
/usr/local/lib/python3.13/dist-packages (from formulaic>=1.1.0-
>statsmodels>=0.14.5->pgmpy) (2.25.0)
Requirement already satisfied: wrapt>=1.17.0rc1 in
/usr/local/lib/python3.13/dist-packages (from formulaic>=1.1.0-
>statsmodels>=0.14.5->pgmpy) (2.4.0)
Requirement already satisfied: anyio in
/usr/local/lib/python3.13/dist-packages (from httpx<1,>=0.23.0-
>huggingface_hub>=0.23->pgmpy) (4.14.2)
Requirement already satisfied: certifi in
/usr/local/lib/python3.13/dist-packages (from httpx<1,>=0.23.0-
>huggingface_hub>=0.23->pgmpy) (2026.7.22)
Requirement already satisfied: httpcore==1.* in
/usr/local/lib/python3.13/dist-packages (from httpx<1,>=0.23.0-
>huggingface_hub>=0.23->pgmpy) (1.0.9)
Requirement already satisfied: idna in /usr/local/lib/python3.13/dist-
packages (from httpx<1,>=0.23.0->huggingface_hub>=0.23->pgmpy) (3.19)
Requirement already satisfied: h11>=0.16 in
/usr/local/lib/python3.13/dist-packages (from httpcore==1.*-
>httpx<1,>=0.23.0->huggingface_hub>=0.23->pgmpy) (0.16.0)
   age  sex  cp  trestbps  chol  fbs  restecg  thalach  exang  oldpeak
slope  \
0   52    1   0       125   212    0        1      168      0      1.0
2
1   53    1   0       140   203    1        0      155      1      3.1
0
2   70    1   0       145   174    0        1      125      1      2.6
```

```
0
3   61    1   0       148   203    0        1      161      0      0.0
2
4   62    0   0       138   294    1        1      106      0      1.9
1

   ca  thal  target
0   2     3       0
1   0     3       0
2   0     3       0
3   1     3       0
4   3     2       0
Learning CPD using Maximum likelihood estimators...
Model Learned Successfully
+-----------+---------------+
| target    |   phi(target) |
+===========+===============+
| target(0) |        0.3365 |
+-----------+---------------+
| target(1) |        0.6635 |
+-----------+---------------+

2. Probability of HeartDisease given evidence= cp
+-----------+---------------+
| target    |   phi(target) |
+===========+===============+
| target(0) |        0.3365 |
+-----------+---------------+
| target(1) |        0.6635 |
+-----------+---------------+
```